

CS:5980

Foundations of Embedded Systems

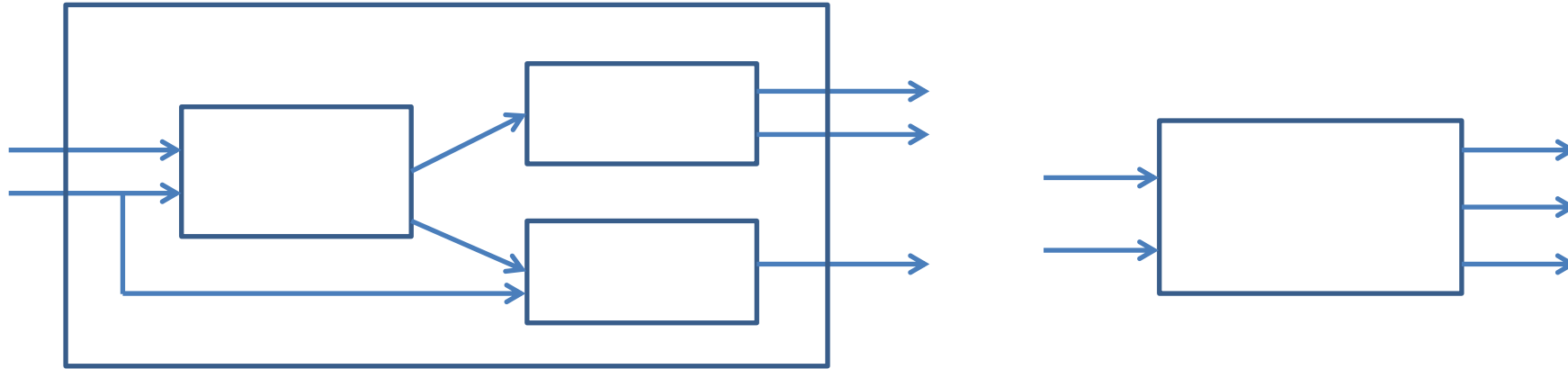
The Synchronous Model

Part I

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Model-Based Design



Block Diagrams

- Widely used in industrial design
- Tools: Simulink, Modelica, LabView, RationalRose, ...

Key question: what is the execution semantics?

- What is a base component?
- How do we compose components to form complex components?

Functional vs. Reactive Computation

Functional model of computation (classical one):

- Given inputs, a program produces outputs
- Desired functionality described by a mathematical function
- **Example:** sorting of names; shortest paths in a weighted graph
- Theory of computation provides foundation
- **Canonical model:** Turing machines



Reactive model of computation:

- System continually interacts with its environment via inputs and outputs
- Desired behavior described by sequences of observed input/output interactions
- **Example:** cruise controller in a car

Sequential vs. Concurrent Computation

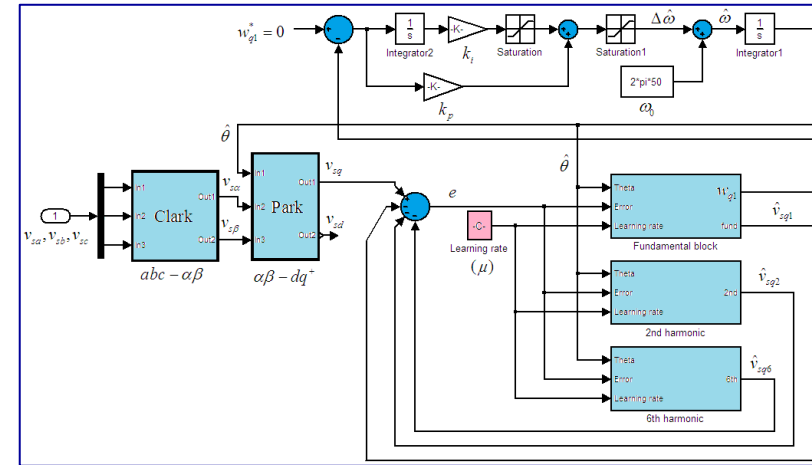
Sequential model of computation (classical one):

- A computation is a sequence of instructions executed one at a time
- Well understood and canonical model: Turing machines

Concurrent model of computation

- Multiple components/processes exchanging information and evolving concurrently
- Logical vs. physical concurrency
- Broad range of formal models for concurrent computation
- **Key distinction:** synchronous vs. asynchronous

Synchronous Models



All components execute in a sequence of (logical) rounds, in lock-step

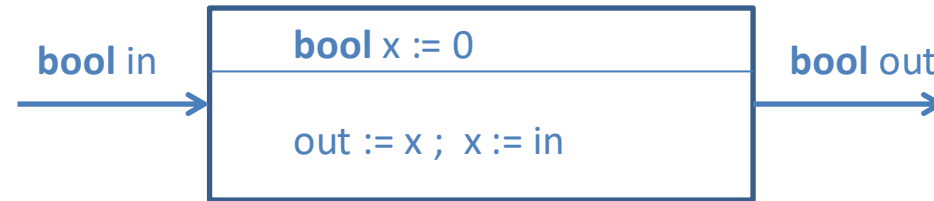
Example: Component blocks in digital hardware circuit

- Clock drives all components in a *synchronized* manner

Key idea in synchronous languages:

- Design system based on such synchronous round-based computation
- **Benefit:** design is simpler (why?)
- **Challenge:** ensure synchronous execution even if implementation platform is not single-chip hardware

First Example: Delay



Input variable: **in** of Boolean type ($\{0,1\}$)

Output variable: **out** of Boolean type

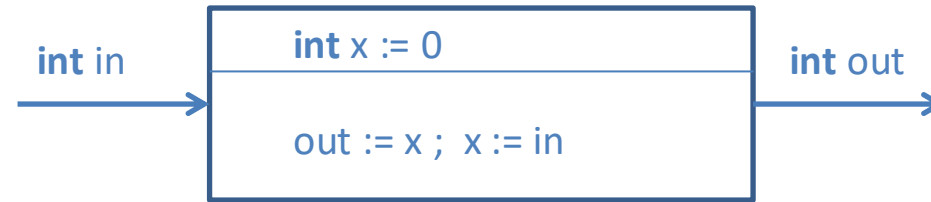
State variable: **x** of Boolean type

Initialization (of state variables): **x := 0**

Execution: in each round, in response to the current input,

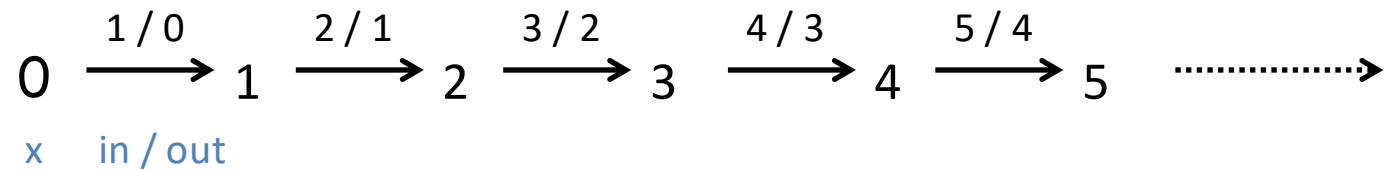
- produce output (**out := x**) and
- update state (**x := in**)

Int Delay: Round-based Execution

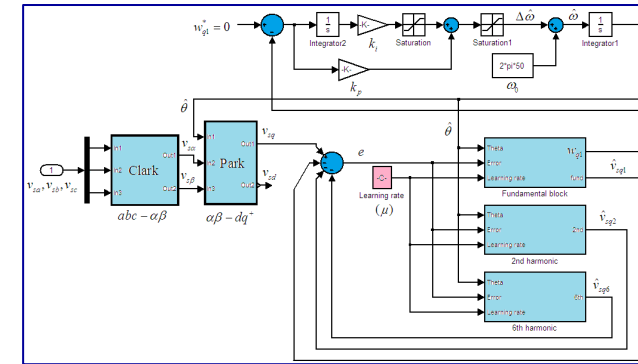


1. Initialize state x to 0
2. Repeatedly execute rounds. In each round:
 - a. Read current value of the input variable in
 - b. Execute the update code to produce output out
 - c. Change state as needed

Example execution:



Synchrony Hypothesis



Assumption: Time needed to execute the update code is negligible compared to delay between successive input arrivals

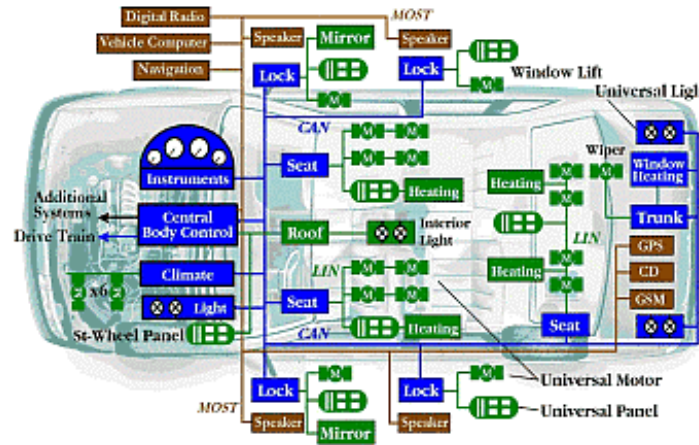
Logical abstraction:

- Execution of update code takes zero time
- Reception of inputs and production of outputs occur simultaneously

Composition: when multiple components are composed, **all** execute synchronously and simultaneously

Implementation: must ensure that this design-time assumption is valid!

Components in an Automobile



Components need to communicate and coordinate over a **shared bus**

Design abstraction: Synchronous **time-triggered communication**

- Time is divided into slots
- In each slot, exactly one component sends a message over the bus

Example: CAN protocol implements time-triggered communication

Model Definition

Syntax: How to describe a component?

- Variable declarations, types, state updates, ...

Semantics: What does the description mean?

- Defined using mathematical concepts such as sets, functions, ...

Formal foundations: Semantics is defined precisely

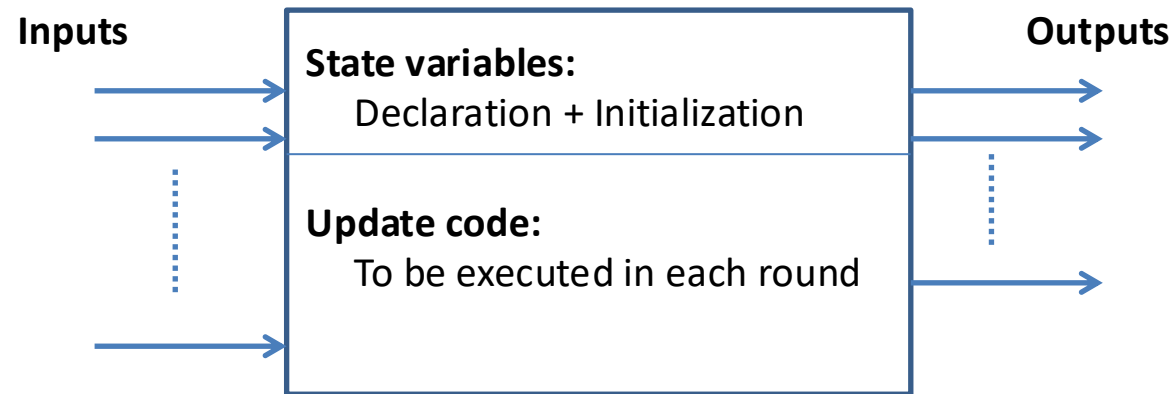
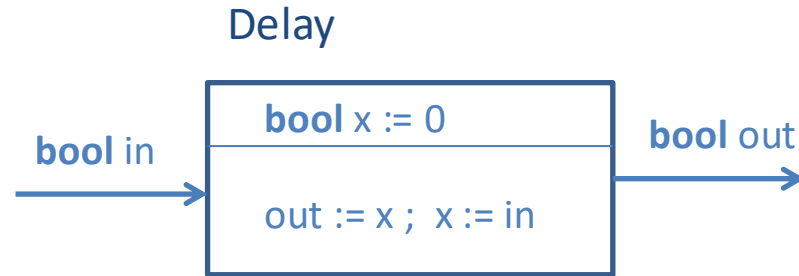
- Necessary for tools for analysis, compilation, verification, ...
- Defining formal semantics for a **real** language is challenging
- But concepts can be illustrated on a **toy** modeling language

Our Modeling Language

Synchronous Reactive Components

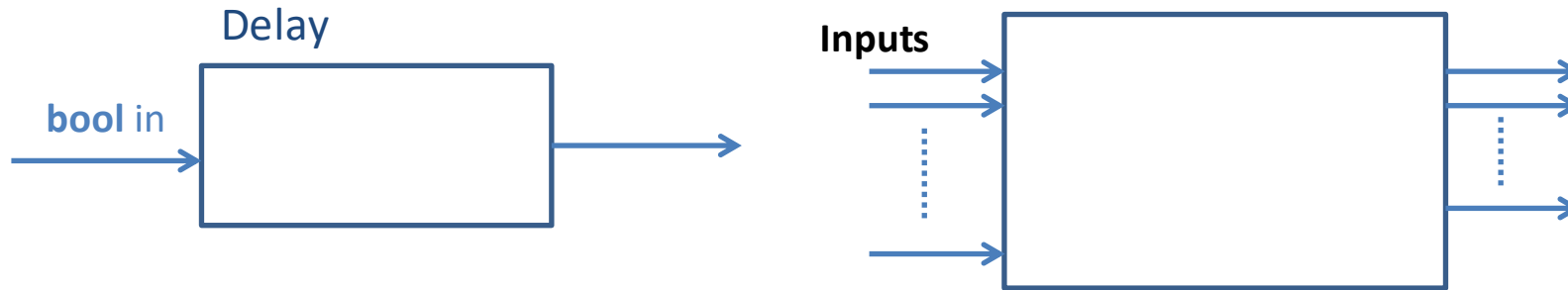
- Representative of many **academic** languages:
Signal, Lustre, Lucid, SMV, Lola, Lucent, ...
- Foundations of **industrial-strength** synchronous languages:
Esterel, SCADE, VHDL, Verilog, Stateflow, ...

Synchronous Reactive Component



Update code contains no function calls or loops

SRC Definition — Inputs



Each component has a set I of input variables

- Variables have types. E.g., **bool**, **int**, **nat**, **real**, {on, off}, ...

Input: Assignment of all the input variables

- The set of possible inputs (input assignments) is Q_I

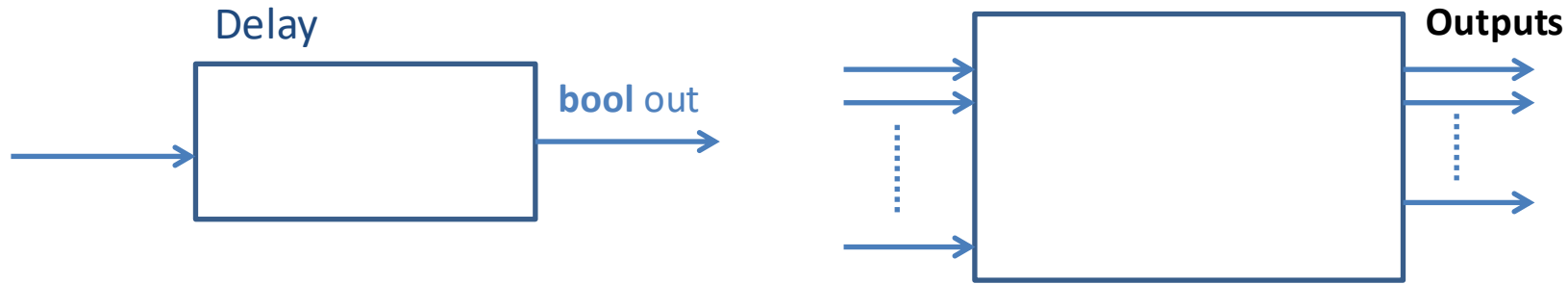
For **Delay**

- I contains a single variable in of type **bool**
- The set of inputs is $\{ \{ \text{in} := 0 \}, \{ \text{in} := 1 \} \}$

Example: I consists of two variables: **int** x , **bool** y

- Each input is a pair: (integer value for x and Boolean value for y)

SRC Definition — Outputs



Each component has a set O of typed output variables

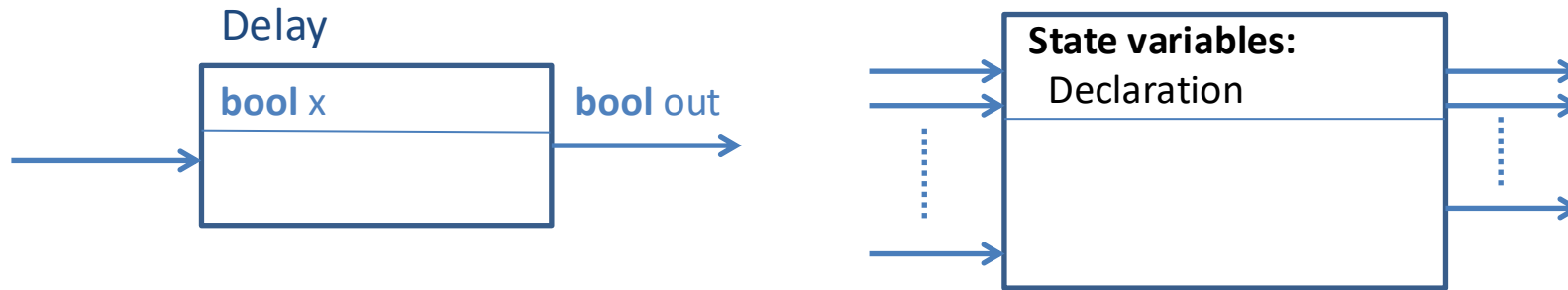
Output: Assignment of all the output variables

- The set of outputs (output assignments) is Q_0

For *Delay*

- O contains a single variable *out* of type **bool**
- The set of outputs is $\{ \{ \text{out} := 0 \}, \{ \text{out} := 1 \} \}$

SRC Definition — States



Each component has a set S of typed state variables

State: Assignment of all the state variables

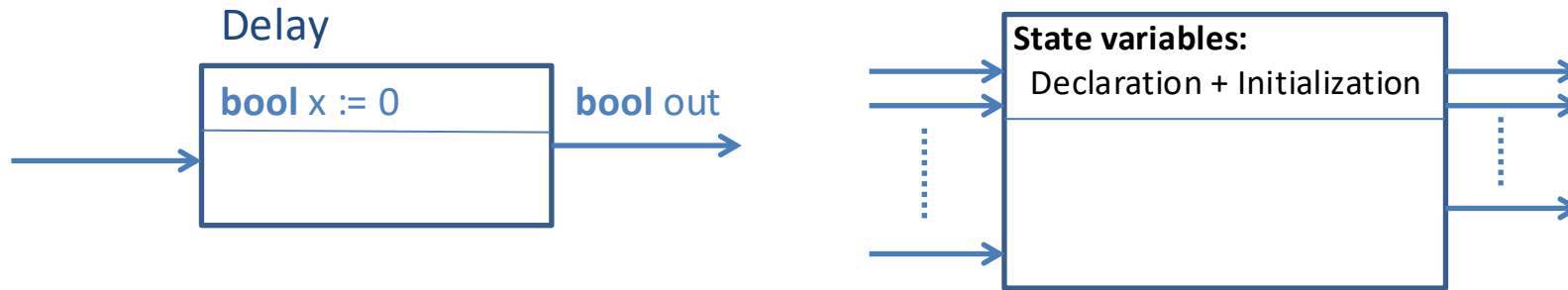
- The set of states is Q_S

For *Delay*

- S contains a single variable x of type **bool**
- The set of states is $\{ \{ x := 0 \}, \{ x := 1 \} \}$

State is internal and maintained across rounds

SRC Definition — Initialization



Initialization of state variables specified by **Init**

- A sequence of assignments to state variables

Semantics of initialization:

- The set $\llbracket \text{Init} \rrbracket$ of initial states, which is a subset of Q

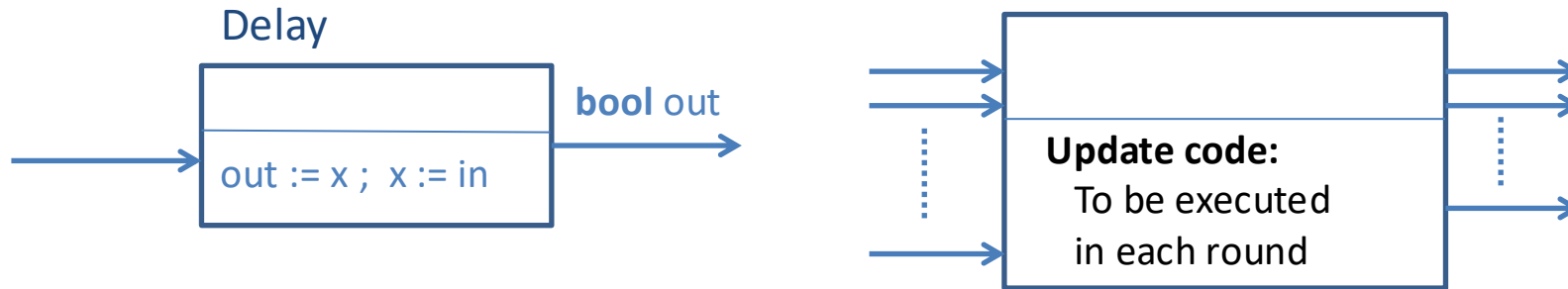
For **Delay**

- **Init** is given by the code fragment $x := 0$
- The set $\llbracket \text{Init} \rrbracket$ of initial states is $\{ \{ x := 0 \} \}$

Components can have multiple (alternative) initial states

- Example: **bool** $x := \text{choose } \{0, 1\}$

SRC Definition — Reactions



Execution in each round given by code fragment **React**

- Sequence of assignments and conditionals that assign output variables and update state variables

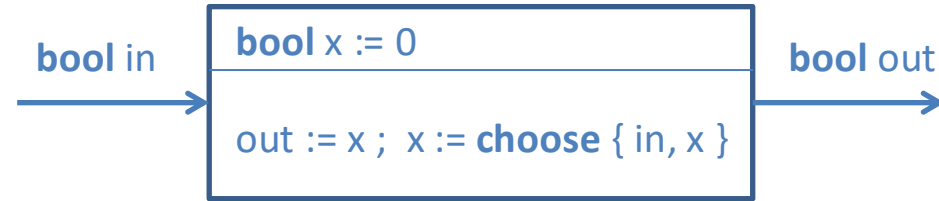
Semantics of update:

- The set $\llbracket \text{React} \rrbracket$ of reactions, where each reaction is of the form:
(old) state – input / output $\rightarrow$ (new) state
- **Note:** $\llbracket \text{React} \rrbracket$ is a subset of $Q_S \times Q_I \times Q_O \times Q_S$

For **Delay**:

- **React** is defined by the code fragment `out := x ; x := in`
- 4 possible reactions: $0 - 0/0 \rightarrow 0$, $0 - 1/0 \rightarrow 1$, $1 - 0/1 \rightarrow 0$, $1 - 1/1 \rightarrow 1$

Multiple Reactions



During update, x is either updated to input in or is left unchanged

- Models the possibility that an input may be **lost or ignored**

Nondeterministic reactions:

- Given (old) state and input, output/new state need not be unique
- The set $\llbracket \text{React} \rrbracket$ of reactions now consists of

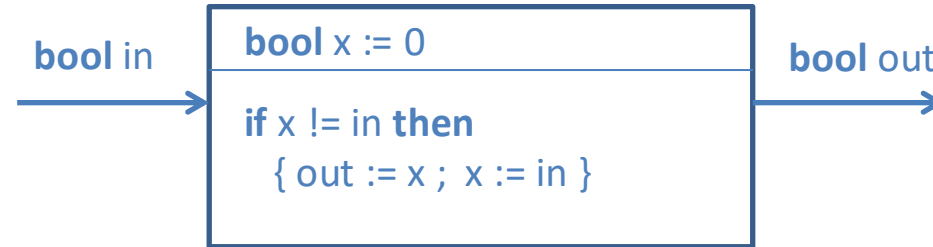
$0 - 0/0 \rightarrow 0$

$0 - 1/0 \rightarrow 1, \quad 0 - 1/0 \rightarrow 0$

$1 - 0/1 \rightarrow 0, \quad 1 - 0/1 \rightarrow 1$

$1 - 1/1 \rightarrow 1$

Selective Reactions



A component may not accept all inputs in all states

- **Motivation:** *blocking* communication

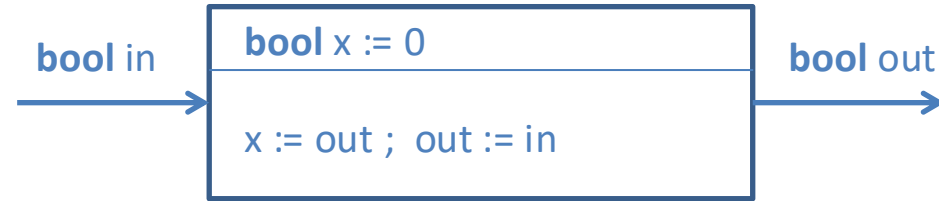
Possible set of reactions in certain state/input combinations may be empty

- The set `[[React]]` of reactions now consists of

$0 - 1/0 \rightarrow 1$

$1 - 0/1 \rightarrow 0$

Syntax Errors



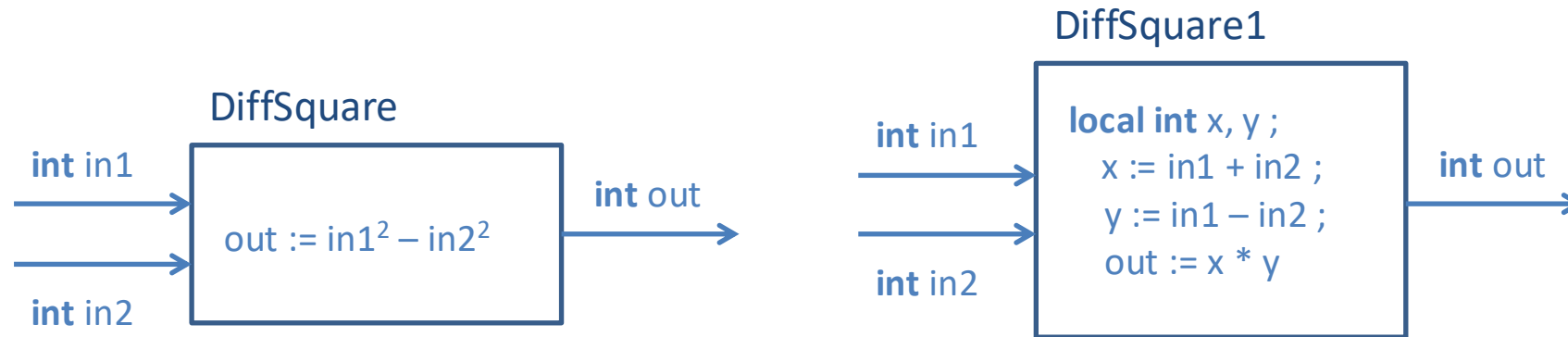
Update code is expected to satisfy a few requirements:

- Types of variables and expressions should match
- **Output** variables must first be **written before** being **read**
- **Output** variables must be explicitly **assigned a value**

Otherwise, no reaction possible

- In above, the set `[[React]]` of reactions is empty

Semantic Equivalence



- These components have identical sets of reactions
- They are syntactically different but semantically equivalent
- Compiler can optimize code as long as semantics is preserved!

Synchronous Reactive Component Definition

- Set I of typed *input variables* gives set Q_I of input assignments, or *inputs*
- Set O of typed *output variables*: gives set Q_O of output assignments, or *outputs*
- Set S of typed *state variables*: gives set Q_S of state assignments, or *states*
- Initialization code *Init*: defines set $\llbracket \text{Init} \rrbracket$ of *initial states*
- Reaction description *React*: defines set $\llbracket \text{React} \rrbracket$ of *reactions* of the form $s - i / o \rightarrow t$, where s, t are states, i is an input, and o is an output

Synchronous languages *in practice*:

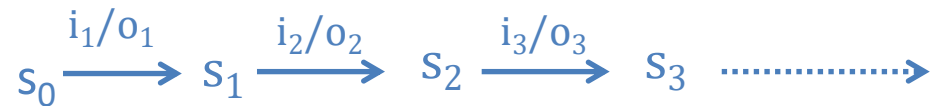
- Richer syntactic features to describe *React*
- Key to understanding: what happens in a single reaction?

Formal semantics: Necessary for development of tools!

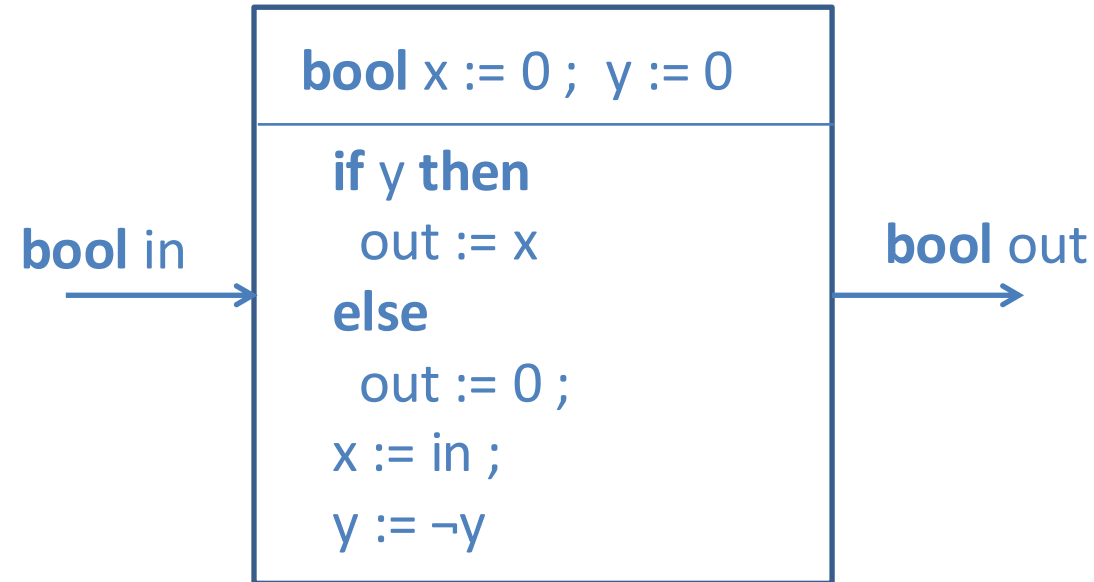
Definition of Executions

1. Given component $C = (I, O, S, \text{Init}, \text{React})$, what are its executions?
2. Initialize **state** to some state s_0 in $\llbracket \text{Init} \rrbracket$
3. Repeatedly execute rounds. In each round $n = 1, 2, 3, \dots$
 - a. Choose an **input** value i_n in Q_I
 - b. Execute **React** to produce **output** o_n and change **state** to s_n
 - that is, $s_{n-1} - i_n / o_n \rightarrow s_n$ must be in $\llbracket \text{React} \rrbracket$

Example execution:



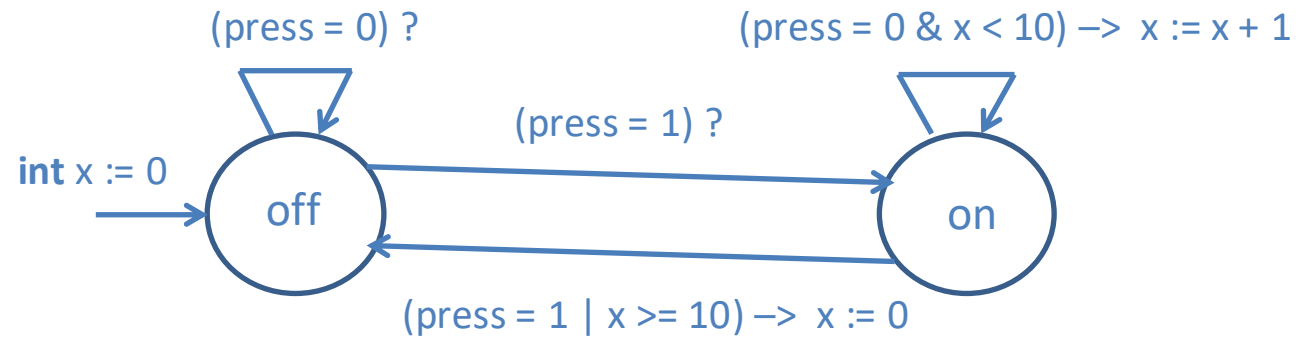
Exercise: What does this component do?



Explain behavior just in terms of the component's input and output

Extended State Machines

Input: **bool** press



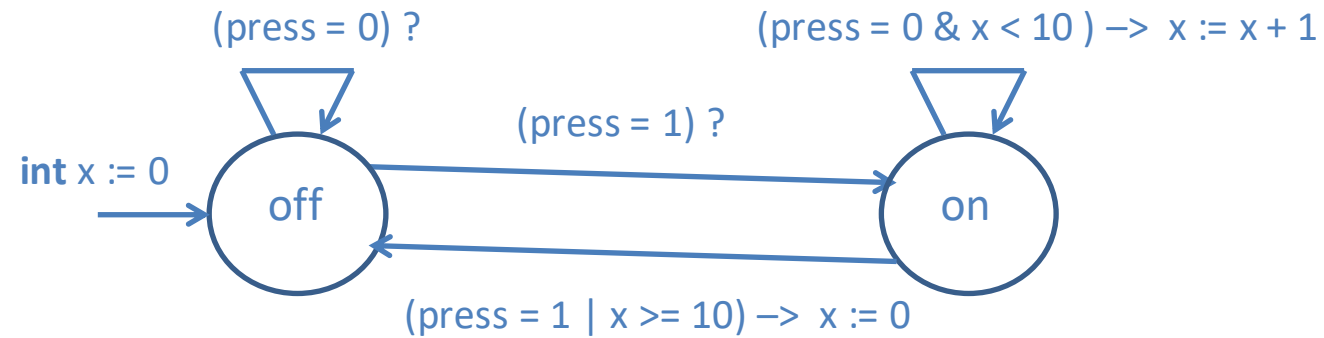
- There is an implicit state variable **mode** ranging over {on, off}
- Reaction corresponds to executing a *mode-switch*

Example mode-switch: from on to off with

guard (press = 0 | x >= 10) and *update* x := 0

Executing ESMs: Switch

Input: **bool** press



State of **Switch** assigns values to **mode** and **x**

Initial state: $(\text{off}, 0)$ (i.e., $\{ \text{mode} := \text{off}, x := 0 \}$)

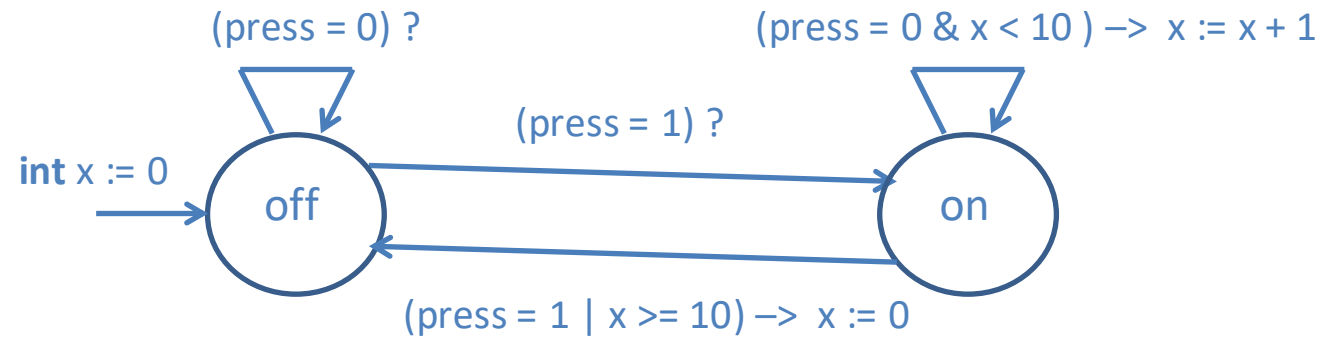
Output: none

Example execution:

$(\text{off}, 0) - 0 \rightarrow (\text{off}, 0) - 1 \rightarrow (\text{on}, 0) - 0 \rightarrow (\text{on}, 1) - 0 \rightarrow (\text{on}, 2) \dots - 0 \rightarrow (\text{on}, 10) - 0 \rightarrow (\text{off}, 0)$

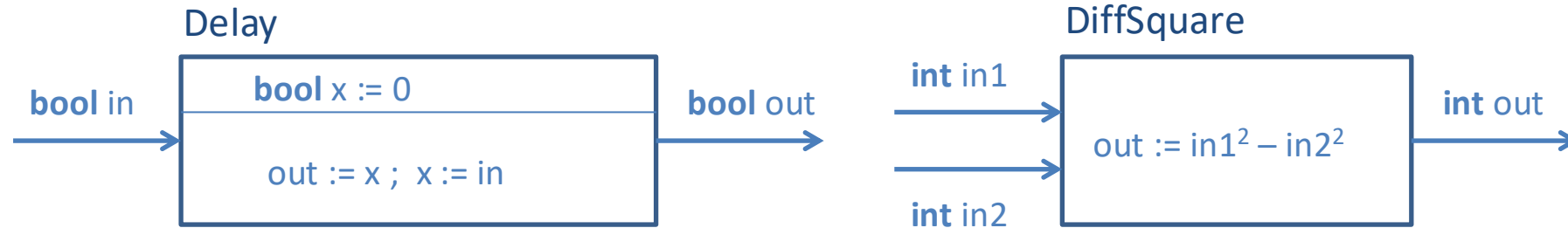
Exercise: ESM to SRC

Input: **bool** press



Rewrite this ESM as a synchronous reactive component

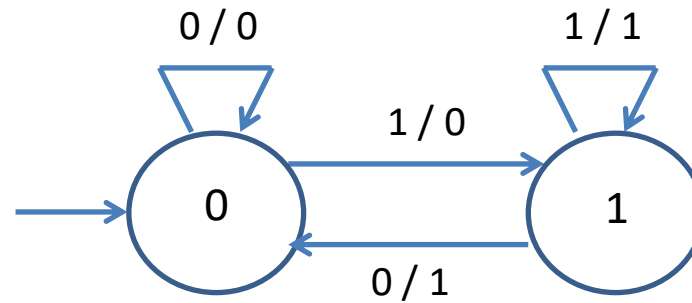
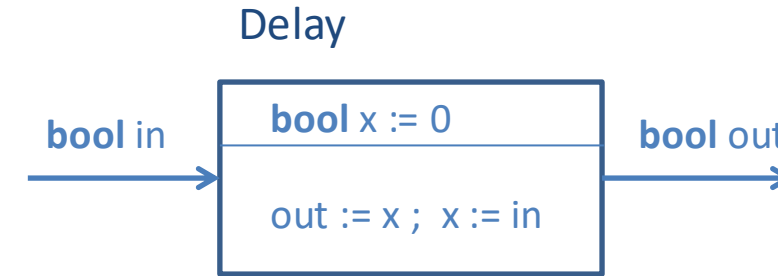
Finite-State Components



A component is *finite-state* if all its variables range over finite types

- *Finite types*: **bool**, enumerated and finite range types (e.g. **{on, off}**, **int[-5..5]**)
- **Delay** is finite-state, but **DiffSquare** is not

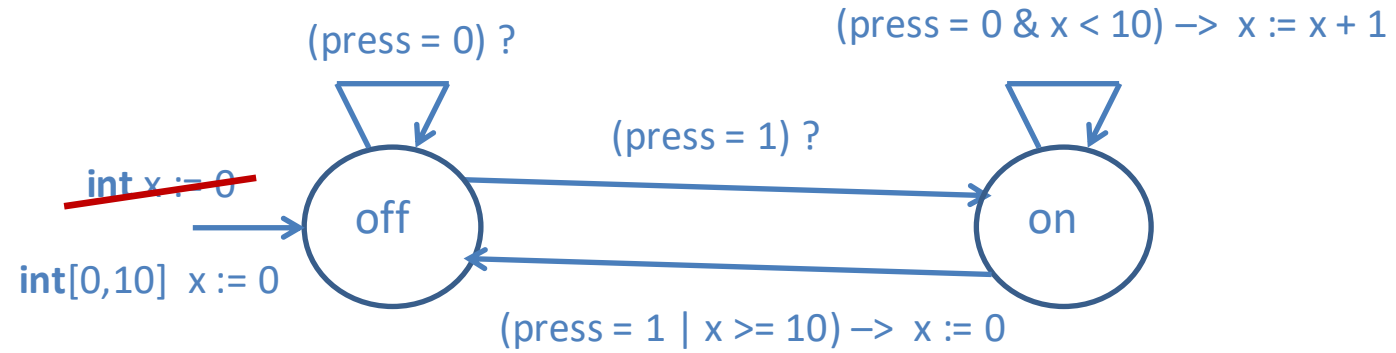
Mealy Machines (for finite-state components)



Finite-state components are amenable to exact, algorithmic analysis

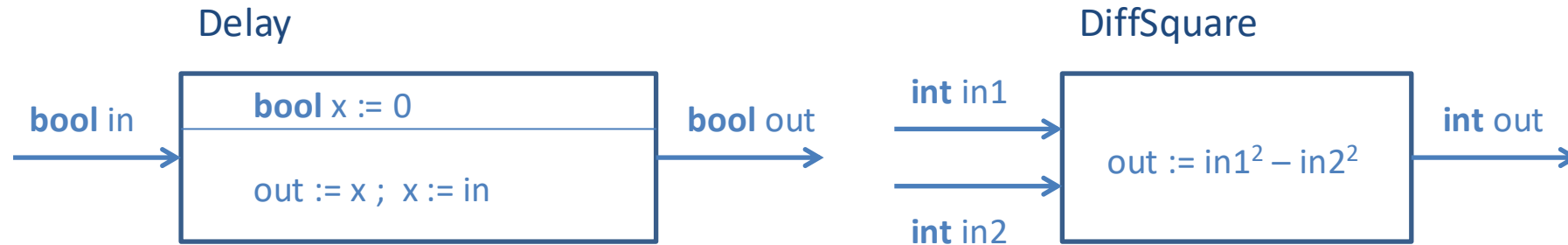
Switch: Is it finite-state?

Input: **bool** press



The system is **effectively** finite-state

Combinational Components



A component is *combinational* if it has **no state** variables

- **DiffSquare** is combinational, but **Delay** is not
- Hardware gates are combinational components

Events

Input/output variable can be of type **event**

An event can be *absent*, or *present*, in which case it has a value

- **event** x means x ranges over $\{\text{present}, \perp\}$
- **event(bool)** x means x ranges over $\{0, 1, \perp\}$
- **event(nat)** x means x ranges over $\{\perp, 0, 1, 2, \dots\}$

Motivation: notion of clock can be different for different components

Syntax:

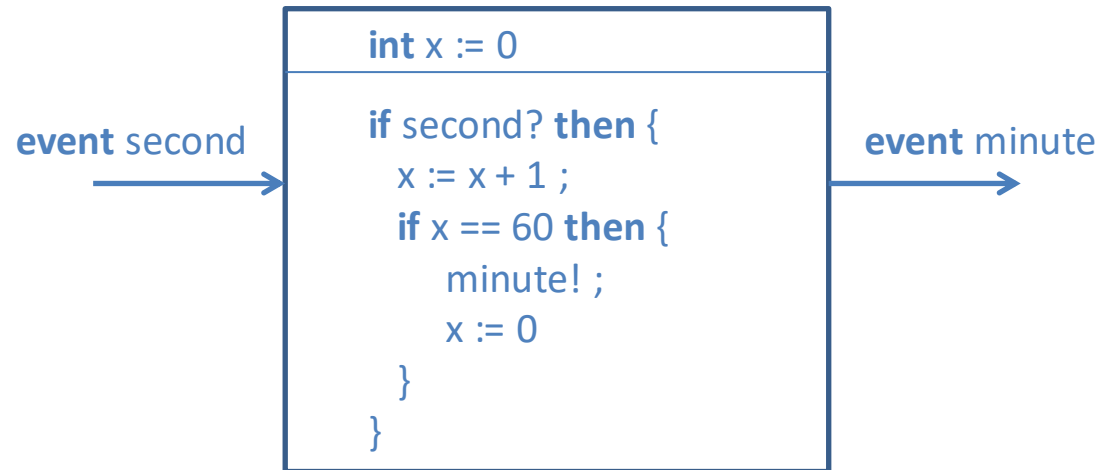
- $x?$ is a short form for the test $x \neq \perp$
- **Syntax:** $x!v$ is a short form for the assignment $x := v$
- **Syntax:** $x!$ is a short form for the assignment $x := \text{present}$ (for **event** x)

Event-based communication:

- If no value is assigned to an output event, then it is *absent by default*
- *Event-triggered component*: executes only in those rounds where input events are present (actual definition slightly more general, see textbook)

Second-To-Minute

Requirement: Issue the output event every 60th time the input event is present

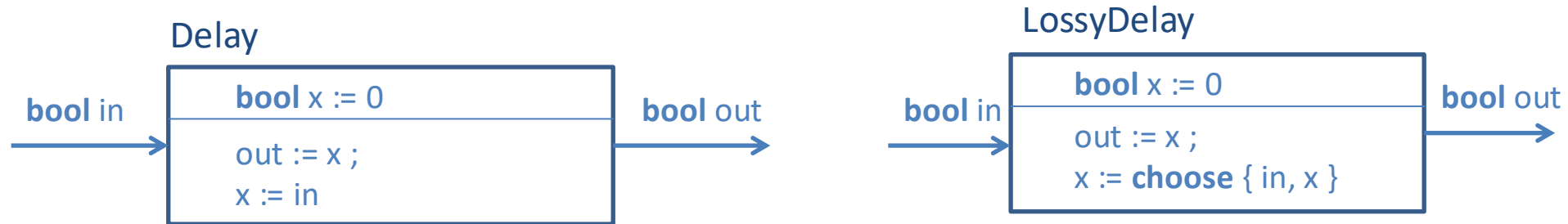


Event-Triggered Components

- Do not execute in a round where triggering input events are absent
- Technically, they do not output any event when inputs are absent

See textbook for formal definition

Deterministic Components

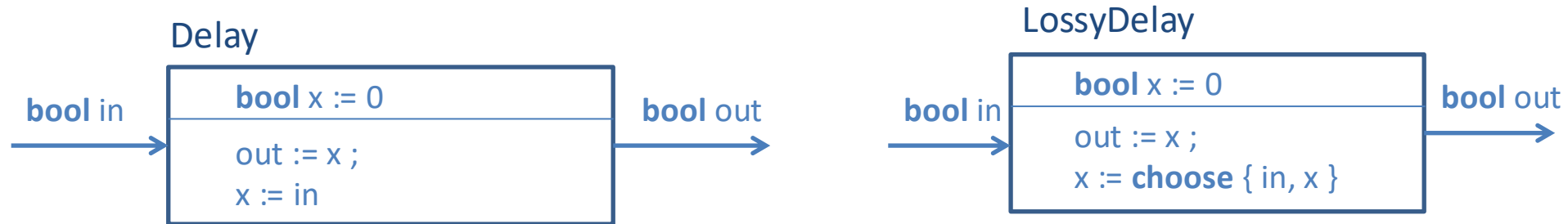


A component is *deterministic* if

1. it has a single initial state, and
2. for every state s and input i , there is a **unique** state t and output o such that $s - i/o \rightarrow t$ is a reaction

Delay is deterministic, but LossyDelay is not

Deterministic Components



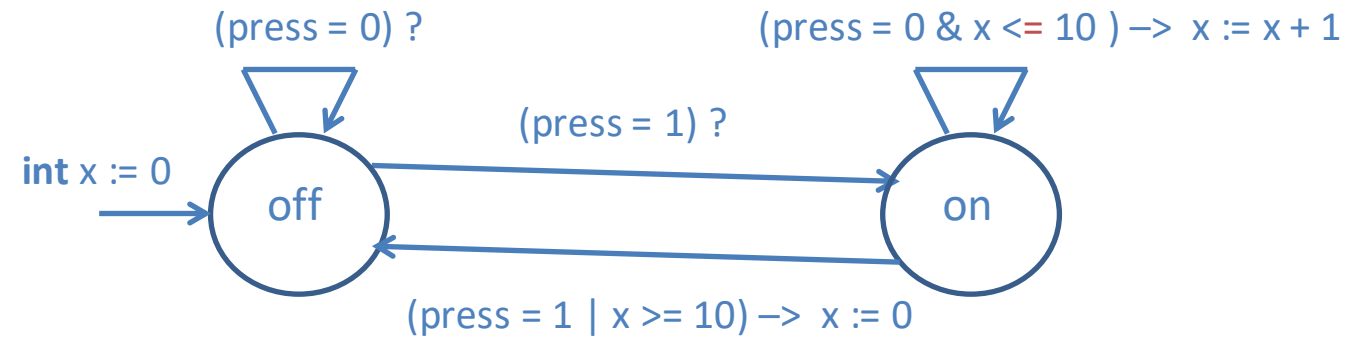
Determinism: same sequence of inputs supplied,
same sequence of outputs observed (predictable, repeatable behavior)

Nondeterminism is

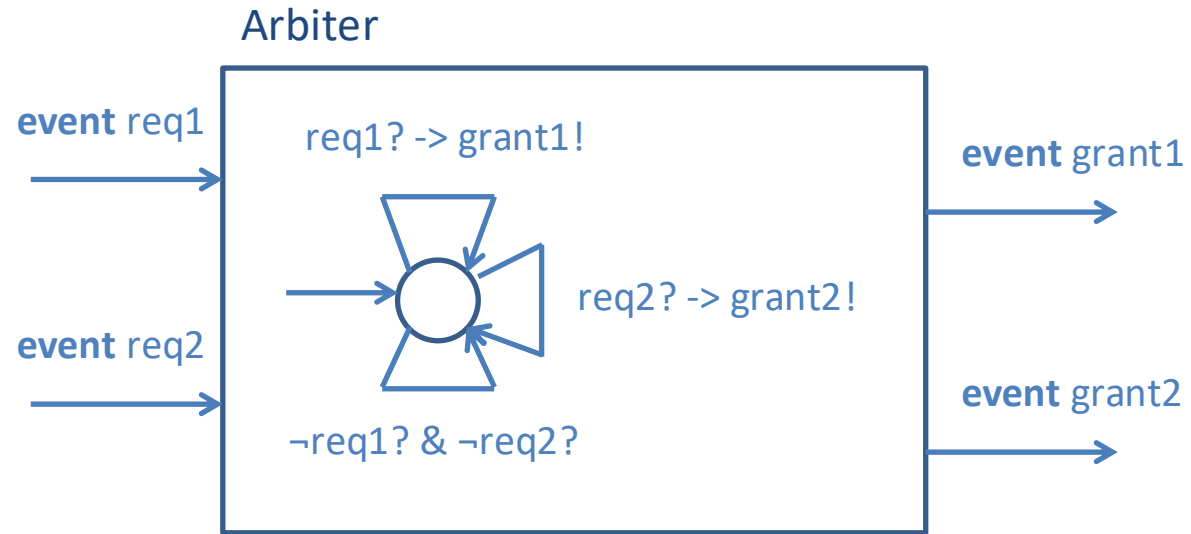
- useful in modeling uncertainty, partial knowledge or abstraction
- different from probabilistic (or random) choice

Modified Switch: What executions are possible?

Input: **bool** press

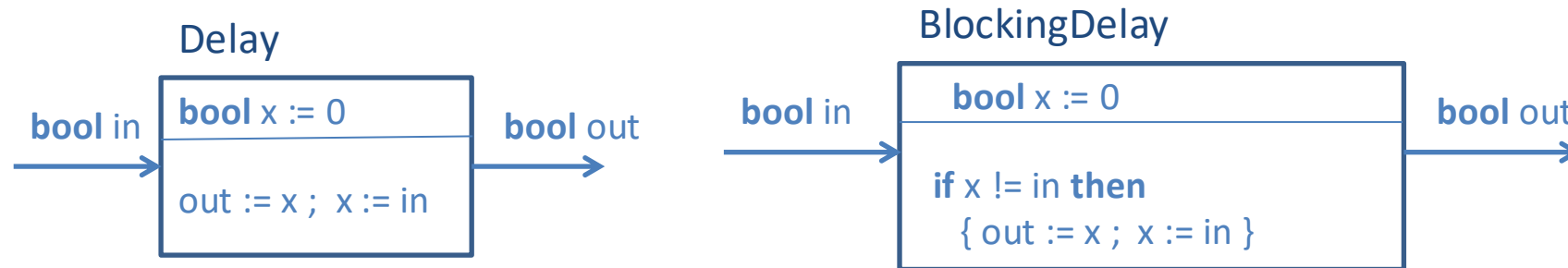


Mixed Notation



What does this component do?

Input-Enabled Components



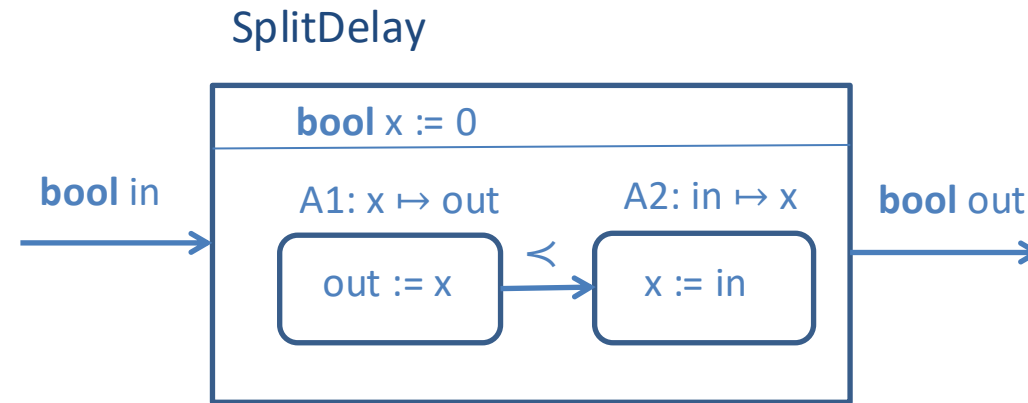
A component is *input-enabled* if for every state s and input i , there is a state t and an output o such that $s - i/o \rightarrow t$ is a reaction

- Delay is input-enabled, but BlockingDelay is not

Non-input-enabled means component is making *assumptions* about the context in which it is going to be used

- When rest of system is designed, we must check that it indeed satisfies those assumptions

Splitting Reaction Code into Tasks



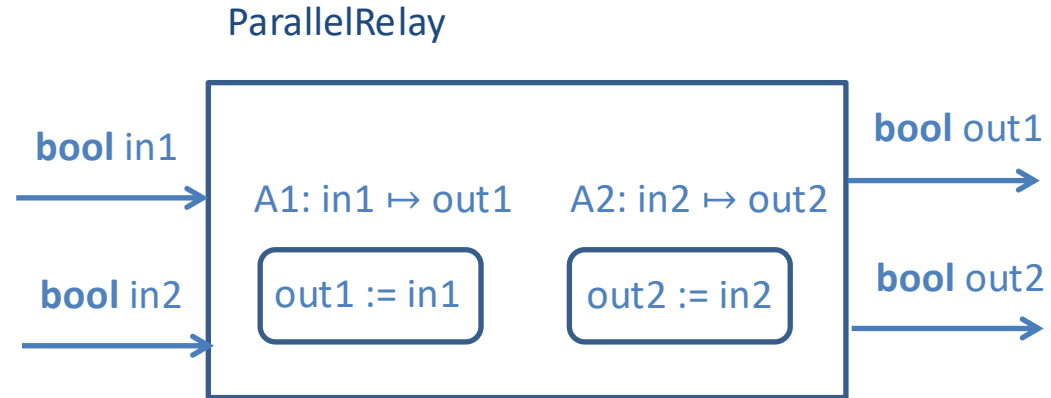
A1 and **A2** are *tasks*: atomic blocks of code

- Each task specifies the variables it reads and writes
- **Example:** **A1** reads **x** and writes **out**

Task Graph: vertices are tasks and edges denote precedence (**<**)

- **A1 < A2** means that **A1** should be executed before **A2**
- Graph must be *acyclic*

Example Task Graph

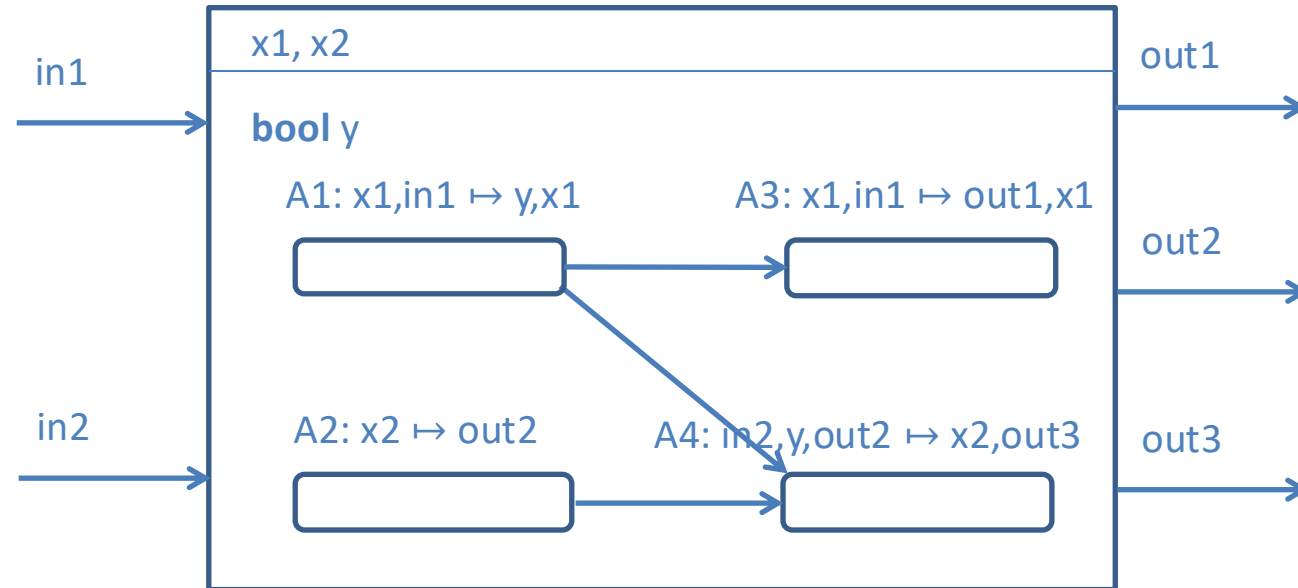


Tasks **A1** and **A2** are unordered

- Possible *schedules* (linear ordering of tasks): **A1**, **A2** and **A2**, **A1**
- All consistent schedules must give the **same** result

I/O *await dependencies*: **out1** awaits **in1**, **out2** awaits **in2**

Example Task Graph



1. What are possible schedules consistent with precedence constraints?
2. What are I/O await dependencies?

Task Graphs: Definition

For a synchronous reactive component C with

- **input** vars I **output** vars O
- **state** vars S **local** vars L

the reaction description is given by

1. a set of tasks, and
2. precedence edges $\prec$ over these tasks

Each task A is specified by:

1. **Read-set** R : a subset of $I \cup S \cup O \cup L$
2. **Write-set** W : a subset of $S \cup O \cup L$
3. **Update**: code to write vars in W based on values of vars in R
 - The set $\llbracket \text{Update} \rrbracket$ is a subset of $Q_R \times Q_W$

Requirements on Task Graph (1)

The precedence relation $<$ must be acyclic

Notation: $A <^+ A'$ means that there is a path from task A to task A' in the task graph using precedence edges

- Relation $<^+$ is the *transitive closure* of the relation $<$

Task schedule: Total ordering $A_1, A_2, \dots, A_n$ of all the tasks that is consistent with the precedence edges

- If $A < A'$, then A must appear before A' in the ordering
- Multiple schedules are possible
- If $A <^+ A'$ then A must appear before A' in *every* schedule

Note: Acyclicity implies that there is at least one task schedule

Requirements on Task Graph (2)

Each output variable is in the write-set of exactly one task

If output y is in write-set of task A , then as soon as A executes, y is available to the rest of the system

If task A writes output y , then y *awaits* an input variable x , written $y \succ x$, if
either the task A reads x
or some another task A' such that $A' \prec^+ A$ reads x

Note: y awaits x means that y cannot be produced before x is supplied

Requirements on Task Graph (3)

Output/local variables are written before being read

If an output or a local variable y is in the read-set of a task A , then y must be in the write-set of some task A' such that $A' \prec^+ A$

Requirements on Task Graph (4)

Tasks with a write conflict must be ordered

There is a *write conflict* between tasks A and A' if a variable written by A is read/written by A'

If A and A' have a write-conflict, the result depends on whether A executes before A' or vice versa

- **Example:** A update is $x := x + 1$; A' update is $out := x$

If tasks A and A' have a write conflict, then they must be ordered: either $A <^+ A'$ or $A' <^+ A$

This way, set of reactions resulting from executing all the tasks do not depend on the task schedule

Task Properties

1. Task $A = (R, W, \text{Update})$ is *deterministic* if for every value $u \in Q_R$ there is a *unique* value $v \in Q_W$ such that $(u, v) \in \llbracket \text{Update} \rrbracket$

If all tasks of a component are deterministic, what can we conclude about the component itself?

2. Task $A = (R, W, \text{Update})$ is *input-enabled* if for every value $u \in Q_R$ there exists at least one value $v \in Q_W$ such that $(u, v) \in \llbracket \text{Update} \rrbracket$

If all tasks of a component are input-enabled, what can we conclude about the component itself?

Credits

Notes based on Chapter 2 of

[Principles of Cyber-Physical Systems](#)

by Rajeev Alur

MIT Press, 2015